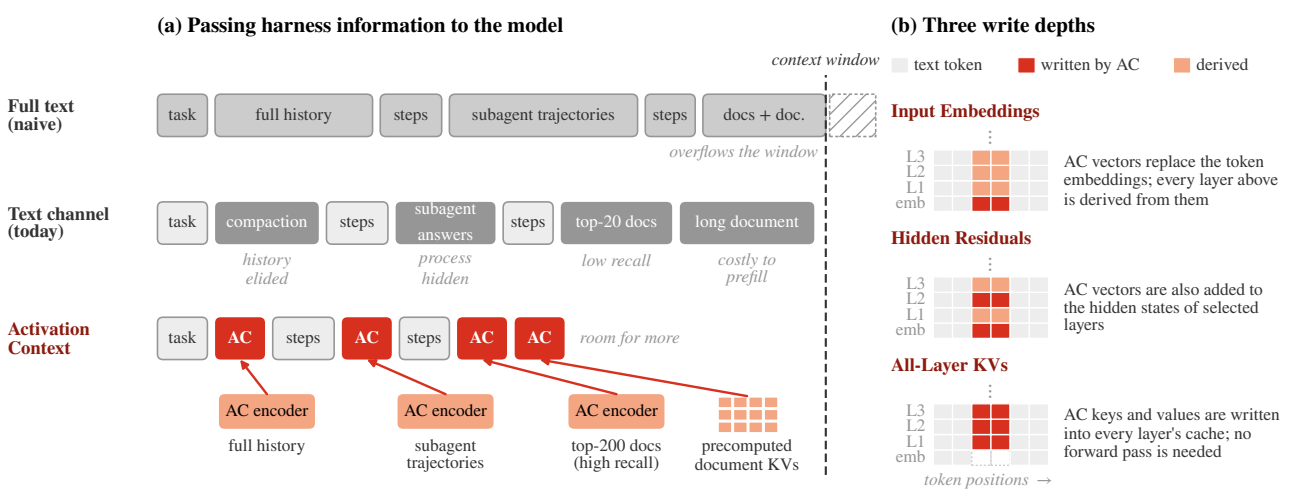
(a) Passing harness information to the model
context window
Full text (naive)
task
full history
steps
subagent trajectories
steps
docs + doc.
overflows the window
Text channel (today)
task
compaction
steps
subagent answers
steps
top-20 docs
long document
history elided
process hidden
low recall
costly to prefill
Activation Context
task
AC
steps
AC
steps
AC
AC
room for more
AC encoder
AC encoder
AC encoder
full history
subagent trajectories
top-200 docs (high recall)
precomputed document KVs
(b) Three write depths
text token
written by AC
derived
Input Embeddings
L3
L2
L1
emb
AC vectors replace the token embeddings; every layer above is derived from them
Hidden Residuals
L3
L2
L1
emb
AC vectors are also added to the hidden states of selected layers
All-Layer KVs
L3
L2
L1
emb
AC keys and values are written into every layer's cache; no forward pass is needed
token positions →